API FUGITIVE EMISSIONS STANDARDS

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WHAT DRIVES STANDARDS

- US EPA REQUIREMENTS ON END USERS TO REDUCE VOLATILE ORGANIC COMPOUNDS (VOC) EMISSIONS
- JUSTICE DEPARTMENT CONSENT DECREES REQUIRE THE USE OF CERTIFIED “LOW EMISSIONS TECHNOLOGY (LOW E)”

LOW-E TECHNOLOGY

- LOW-E VALVE IS DEFINED AS: A valve (including its specific packing assembly or stem sealing component) for which the manufacturer has issued a written warranty that it will not emit fugitives at greater than 100 PPMv, and that, if it does so emit at any time in the first 5 years, the manufacturer will replace the valve; provided, however, that no valve shall qualify as “Low-E” by reason of written warranty unless the valve (including its specific packing assemble):

LOW-E VALVE TECHNOLOGY

- CRITERIA
  - First was tested by a manufacturer or qualified testing firm pursuant to generally-accepted good engineering practices for testing fugitive emissions; or
  - Is an “extension” of another valve that qualified as “Low-E”
- Best Practices for Low-E Valve Technology
  - Upon installation, a facility must check that the packing gland nut or packing torque is at the manufacturers recommendation; and
  - Within 2 weeks of installing the Low-E Valve technology, a facility must check that the packing gland nut or packing torque is at the manufacturers recommendation

EPA CONSENT DECREE NEW EQUIPMENT UPDGRADE REQUIREMENTS

- Newly installed valves must be Low-E
- Repack or replace with Low-E for valves that leak above 250 PPMv (Chemical)
- Repack or replace with Low-E for valves that leak above 500 PPMv (Refinery)

API FUGITIVE EMISSIONS STANDARDS

- API 622 2ND EDITION- “TYPE TESTING OF PROCESS VALVE PACKING for FUGITIVE EMISSIONS”
- API 624 1ST EDITION- “TYPE TESTING of RISING STEM VALVES EQUIPPED with GRAPHITIC PACKING for FUGITIVE EMISSIONS
- API 641 1ST EDITION *(IN DEVELOPMENT)*- “TYPE TESTING of QUARTER-TURN VALVES for FUGITIVE EMISSIONS”

API 622 2ND EDITION
"TYPE TESTING OF PROCESS VALVE PACKING for FUGITIVE EMISSIONS"

- Sets Requirements & Parameters for following tests:
  - Emissions
  - Corrosion
  - Packing material composition and components
- Graphitic Packing Only
- Test Methods Apply Only to:
  - Rising stem
  - Rotating stem

API 622 2ND EDITION
Emissions Test

- EPA Method 21
- Methane Test Media Only
- 1510 Mechanical Cycles
- 5 Thermal Cycles- Ambient to 500° F
- Allows for 1 Re-torque of Packing Gland During Test
- Testing is Discontinued if Leakage Exceeds 500 PPMv
- Static Leak Measurements Only
- Optional Low Temperature Test at -20° F Allowed at Completion of High Temp Test
- Defined Stroke Speed of Test

API 622 2nd EDITION

- Testing is done in a test rig that utilizes B7 studs for tightening packing gland follower
- This has prevented a possible problem for testing the higher density graphite packing that is being manufactured to meet the lower emissions ratings from the EPA
- The higher density graphite packing requires higher loading torques than those previously used in valve service

API 622 2ND EDITION
Corrosion Test

- Evaluation of “hot” and “cold” corrosion caused by packing.
- Provides means for evaluating the effect of inhibitor systems and valve stem metallurgy combinations with respect to corrosion rate and weight loss.
- Information required before performing test:
  - Primary material used in manufacture
  - Type of corrosion inhibitor(s)
  - Inhibitor content by weight
  - Method of application/distribution of inhibitors in/on the packing (ie. active or passive)
  - Packing test specimen size identical to that required for leak testing

API 622 2ND EDITION

Packing Material and Components Test

- Weight Loss
- Density
- Lubricant content
- Leachable

API622 3rd EDITION CHANGES

- API 622 2nd Edition was published in 2011
- New Task Force created to begin working on revisions Spring 2014 with expected republication 2016.
- Proposed Changes to current edition
  - Pass Fail Criteria for leak rate < 100 PPMv
  - Testing for Elastomeric Seals
  - Testing for TFE based configurations
  - Addition of Dynamic Leak Measurements
  - One Re-torque?
- Joint work team with Upstream to help develop a standard they would be able to use

API 624 1ST EDITION

TYPE TESTING OF RISING STEM VALVES with GRAPHITE PACKING for FUGITIVE EMISSIONS

- Rising and Rising-Rotating Stem Valves
- Graphite Packing that Successfully Completed API 622 2nd Edition Only
- Methane Test Media Only
- 310 Mechanical Cycles
- 3 Thermal Cycles- Ambient to 500° F

API 624 1ST EDITION

- Static and Dynamic Leak Measurements taken and recorded for test
- No Re-torque of Packing Gland Allowed
- Optional Low Temperature Test at -20° F Allowed at Completion of High Temp Test
- Valve Size Selections Qualify Product Range

API 624 1st EDITION CHALLENGES

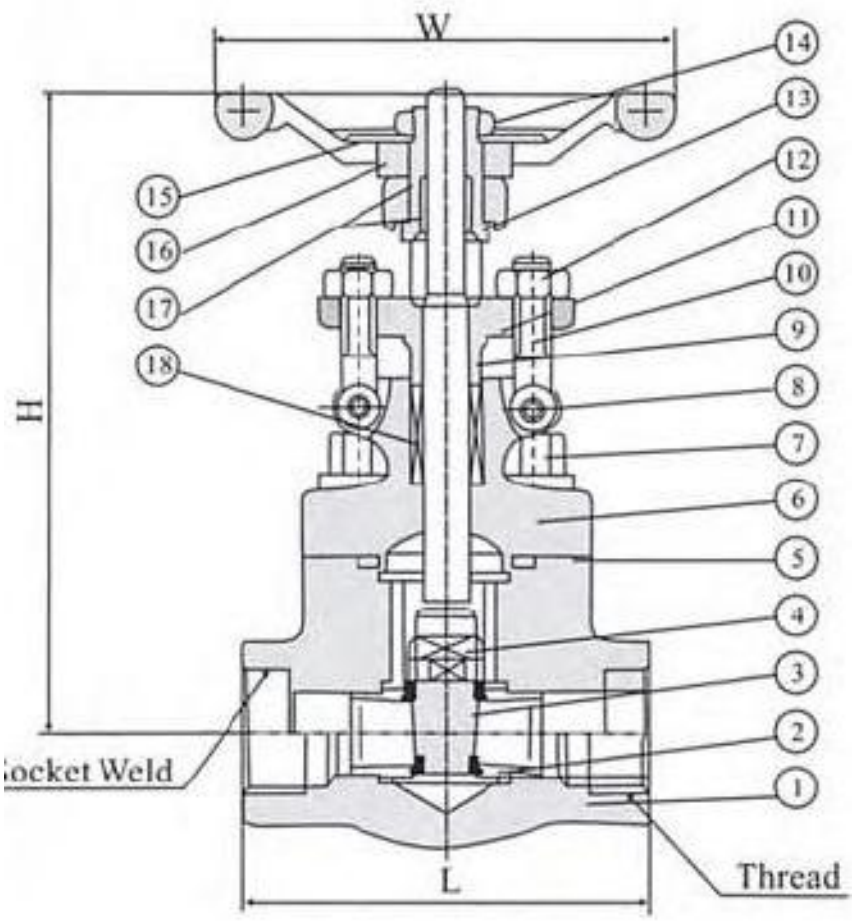
- Stem is oriented in vertical position which can allow yoke bushing grease to travel down stem to packing area.
- B7 studs is the standard for the test rig used in API 622 2nd Edition. If not B7 there might not be enough bolt torque to get the initial seat required for the higher density braided packing that have successfully passed 622.
- Selection of Proper Stem & Stud Grease Becomes Critical due to Thermal Testing Parameters.

API 624 1st EDITION CHALLENGES

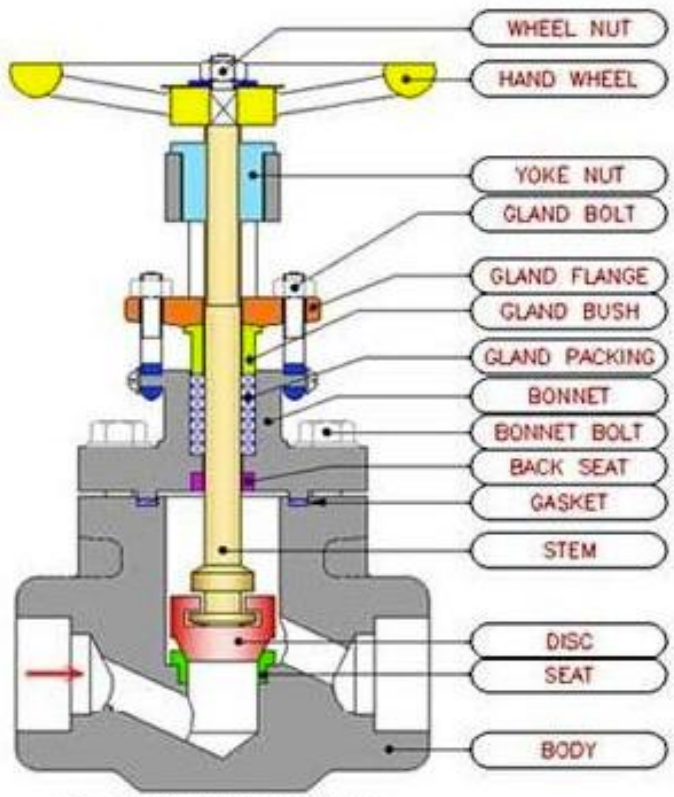
- Commodity API600 Valves to meet LE expectations
- Strength of mechanical components need to be much higher.
- Stem & Bore tolerances need to be held to much tighter requirements.
- Mechanical stroke speeds are critical to not damaging the mechanical components. This leads to longer test completion times due to the variances from valve sizes.

API 624 1st EDITION CHALLENGES

- Stem
  - Roughness
  - Hardness
  - Thread Profile
  - Thread Fit tolerance to Stem Nut
- Packing Gland Flange
  - Strength
  - Hardness
  - Surface Roughness
  - Clearances to Stem



API 624 1st EDITION CHALLENGES

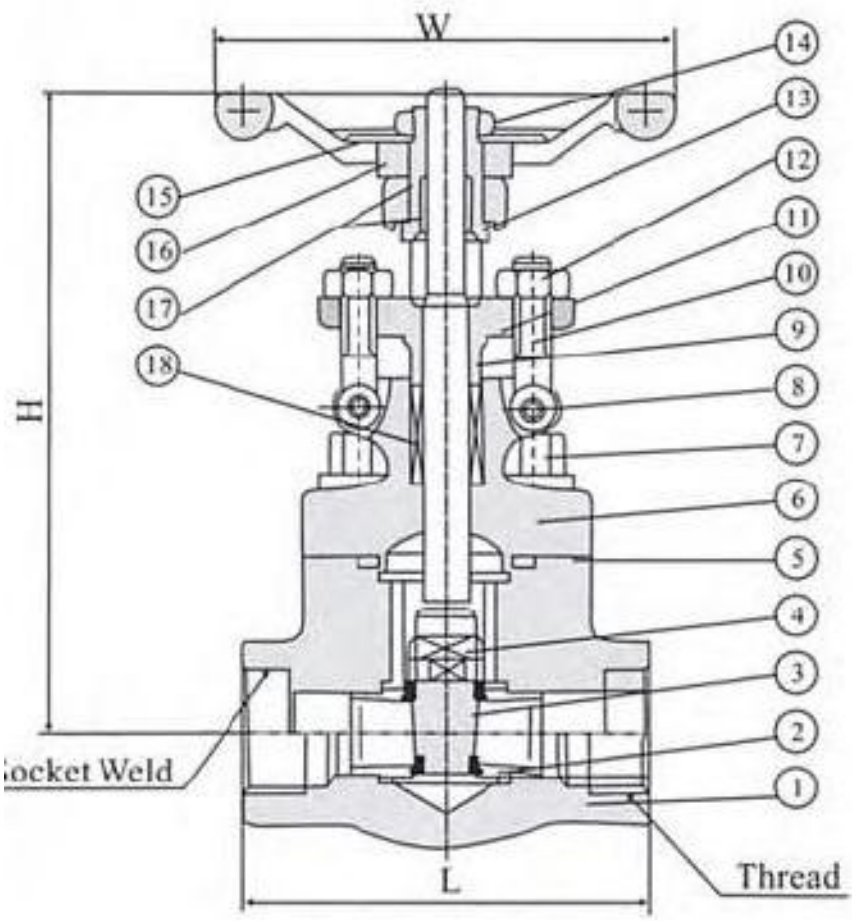


Forged Steel Globe Valve

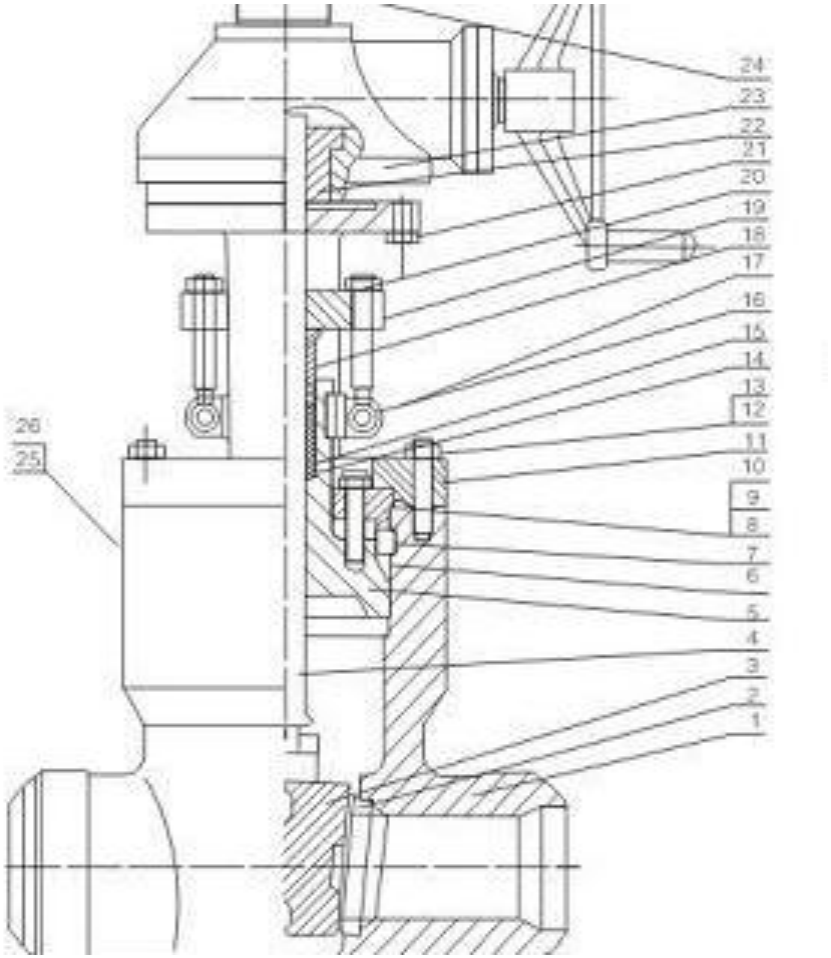
- Stuffing Box
  - Flatness
  - Roughness
  - Tolerance Control
- Stem Nut
  - Thread Profile
  - Thread Fit with Stem Threads
  - Tight Control of Tolerances
  - Define Allowable Clearance

API 624 1st EDITION CHALLENGES

- Packing Gland Bolt & Nut
  - B7 w/ 2H Nuts???
  - B8M if stainless???
  - Hardened Washers???
- Packing Installation
  - Higher Torque to Attain Initial Seat
  - Tighter Clearance Requirements Between Stem & Packing Gland Than API600 Requires



API 624 1st EDITION CHALLENGES



- Gear Operated Valves
  - Remove Grease Before Test as Safety Precaution
  - Increases Time of Test Due to Stroke Speed Limitations
  - Additional Rigging Time for Test Setup

API 624 1st EDITION CHALLENGES

- So You pass the test and start selling API624 valves. What happens when it comes times to be installed?
  - Lubrication
  - Packing Gland Torque is checked to see it meets original required factory setting. EPA Consent Decree requires checking within 2 weeks of installation to verify it meets manufacturer setting.
  - Visual Inspection for any damages from shipping

API 624 1st EDITION CHALLENGES

- Repair Considerations once installed:
  - Factory Certified Repair facility or Field Repair
  - API RP621 followed by both repair facility and field repair team?
  - Packing Manufacturer Certified Installer Training?
  - Issuance of 5 year warranty after repair by Manufacturer or repair facility/field repair?

API 624 2nd Edition Considerations

- API SCOPV would begin task group to develop 2nd Edition in 2017.
- Possible considerations for next edition:
  - Is the intent of API624 to test the mechanical integrity of the valve or to test emissions level of a packing within a valve design?
  - More similarities to API622:
    - Cycle Speeds Defined in API622 but not in API624
    - API622 allows the system to be pressured down at the end of each day, while API624 does not.

API 624 2nd Edition Considerations

- Completing the testing takes significant time and resources. Stuffing box design remains the same over many size ranges and pressure classes. Is there a possibility to reduce the number of valves to be tested?
- Many globe valve manufacturers for LE are beginning to design to the non-rotating stem. Is it possible to reduce the number of globe valves to test if it is utilizing the same seal system as the gate valve to qualify size & pressure class for both?

API 641 1ST EDITION

TYPE TESTING of QUARTER-TURN VALVES for FUGITIVE EMISSIONS

- Is currently in ballot with results anticipated for Spring 2015 meeting discussions.
- Graphite Packing that successfully completed API 622 2nd Edition only.
- Allows TFE and elastomeric sealing designs to be tested and qualified.
- 1510 Mechanical Cycles ???

API 641 1ST EDITION

- 3 thermal cycles- Ambient to 500° F
- Optional low temperature test at -20° F allowed at completion of high temperature thermal cycle test.
- Discussions on adding true cryogenic low temperature test to standard to meet growing LNG requirements.
- Valve size selections qualify product range.

QUESTIONS?

THANK YOU!

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